
Probing-Based Test-Time Steering of Music Diffusion Transformers

Junyoung Koh^{1 2 3}

Abstract

Pretrained music diffusion transformers expose text as their primary control surface, making fine-grained musical content difficult to manipulate. Studying Stable Audio Open’s DiT, we show that simple linear probes on intermediate activations recover strong concept directions for instruments, vocals, and genre. Treating these probe coefficients as concept activation vectors and injecting them into the DiT during generation steers probe-measured concept presence monotonically, exhibits diagonal-dominant effects in concept space, transfers across diverse base prompts, and is strongest at early layers. An independent audio-language model corroborates concept suppression and suggests that moderate positive steering can add target concepts in some settings, while overly strong steering can degrade the generation. Linear probing thus gives a simple handle for interpreting and controlling music diffusion transformers beyond the text prompt, turning their internal concept representations into a usable one-parameter steering knob that fine-tunes what the text condition already permits rather than rewriting it.

1. Introduction

Pretrained text-conditioned music generation models such as Stable Audio Open (Evans et al., 2024), DiffRhythm (Ning et al., 2024), MusicGen (Copet et al., 2023), and MusicLM (Agostinelli et al., 2023a) produce high-quality controllable music from text, yet their internal mechanisms remain poorly understood. Text prompts are also a coarse control surface that does not expose fine-grained operations such as adding drums without changing genre, or removing synth bass without altering vocals.

We ask what information the model carries internally and

¹Department of Artificial Intelligence, Yonsei University, Republic of Korea ²MAAP Lab ³KRAFTON. Correspondence to: Junyoung Koh <solbon1212@yonsei.ac.kr>.

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whether it can be turned into a control surface beyond the text prompt. We probe intermediate activations of Stable Audio Open’s diffusion transformer for eleven instrument and genre concepts using simple linear classifiers, and find that concept information is broadly available and linearly decodable, with the strongest signal on instruments. We then treat the probe coefficients as concept activation vectors (Kim et al., 2018) and inject them additively into a single DiT layer during generation, obtaining a one-parameter knob that modulates probe-measured concept presence in the generated audio. Mechanistic interpretability of pretrained music diffusion transformers is still largely unexplored, in contrast to the rich probing and steering literature for language models. Our contributions are threefold:

- Linear probes on intermediate activations of Stable Audio Open’s DiT recover strong concept directions for instruments, vocals, and several genres.
- Treating these probe coefficients as concept activation vectors and injecting them into the DiT during generation enables monotonic, diagonal-dominant test-time steering of probe-measured concept presence in the output, strongest at earlier layers and robust across diverse base prompts.
- An independent audio-language model corroborates the audio-side effect, with negative steering consistently suppressing the target concept and moderate positive steering able to increase concept similarity in some settings before stronger interventions degrade audio.

2. Related Work

Concept activation vectors and steering. The idea that concepts occupy linear directions in activation space goes back to TCAV (Kim et al., 2018), which reads concept use off a linear classifier. This view has since become a control tool, as representation engineering (Zou et al., 2023) steers LLM safety and persona and additive interventions elicit truthful or otherwise modified behavior (Li et al., 2023; Turner et al., 2024). Such directions build on linear probing, a long-standing way to interpret frozen networks (Alain & Bengio, 2017). A complementary tradition instead decomposes activations into sparse dictionaries (Olshausen &

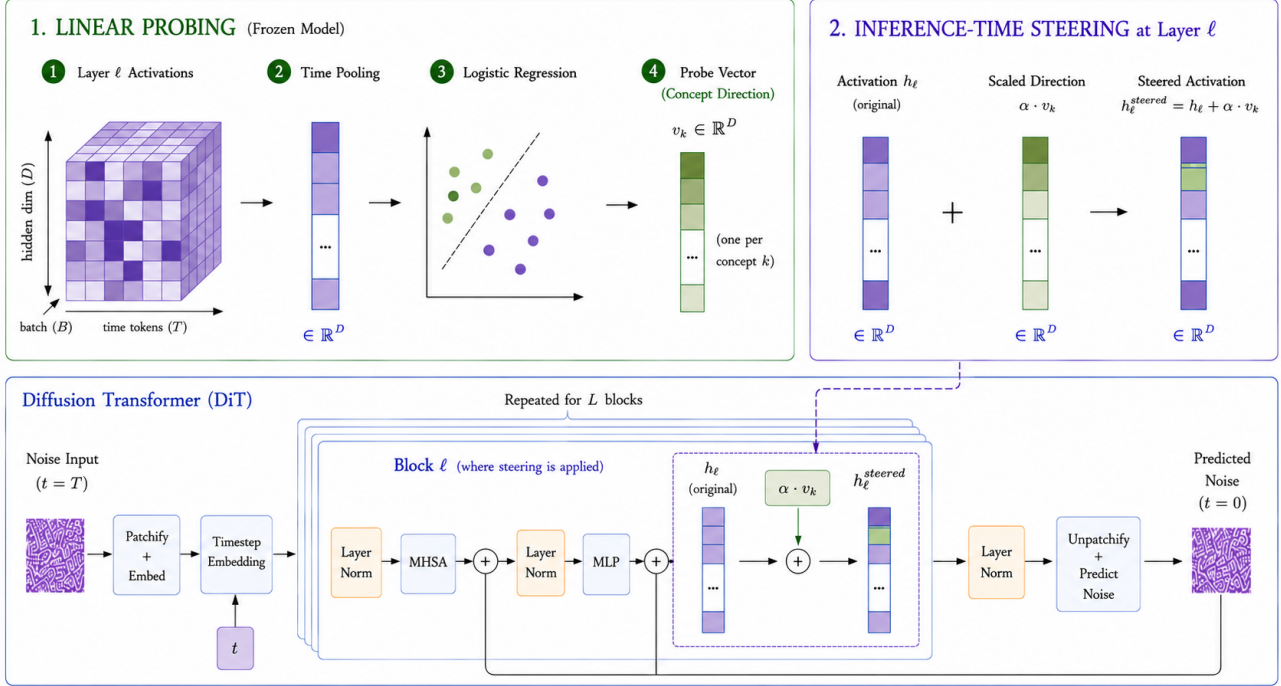


Figure 1. Overview of our framework. (1) Activations from layer ℓ of a frozen Stable Audio Open DiT are time-pooled and probed with class-balanced logistic regression, yielding a Concept Activation Vector $v_k \in \mathbb{R}^D$ per concept, which is ℓ_2 -normalized to $\tilde{v}_k$ for steering. (2) At inference the scaled direction $\alpha \tilde{v}_k$ is added to the activation, $\hat{h}_\ell \leftarrow h_\ell + \alpha \tilde{v}_k$. (3) The injection is applied inside DiT block ℓ at every diffusion step, and the causal effect is measured by re-probing the output and by an independent CLAP check.

Field, 1996; Bricken et al., 2023; Templeton et al., 2024; Gao et al., 2024). Both lines of work were developed almost entirely on language models, and how they transfer to a generative *music* DiT is largely open.

Audio probing and music editing. Probes localize concept information in pretrained audio encoders such as MERT (Li et al., 2024), Wav2Vec2 (Baevski et al., 2020), and CLAP (Wu et al., 2023), but these analyses target encoders rather than generative DiTs and do not study steerability. On the generation side, a range of methods edit music by acting on the text condition or latent state, including MusicMagus (Zhang et al., 2024), AudioLDM editing (Liu et al., 2023a; 2024), and SDEdit (Meng et al., 2022) or DDIM-inversion (Song et al., 2021) variants, while instrument-level control has been pursued through attention-band alignment (Koh et al., 2025) and auxiliary conditioning branches (Koh, 2026). What remains comparatively unexplored is whether the DiT’s own internal activations expose concept directions that can be manipulated directly.

3. Method

Setup. Figure 1 gives an overview, and the full experimental configuration is tabulated in the supplementary material. Stable Audio Open (Evans et al., 2024) is a text-conditioned

music diffusion model that pairs an Ooblick-style VAE pre-transform with downsampling ratio $1920\times$ and a rectified-flow (Liu et al., 2023b) diffusion transformer (Peebles & Xie, 2023; Vaswani et al., 2017).

Activation collection. Stable Audio Open’s DiT is a stack of $L = 24$ pre-norm transformer blocks, each operating at hidden dimension $D = 1536$. It denoises a VAE latent $z_t \in \mathbb{R}^{C \times T}$ with $C = 64$ channels and $T = 216$ time frames (the tokens of the transformer) for a ten-second clip. We build a corpus of $N = 150$ such clips, comprising 100 MUS-ICCAPS (Agostinelli et al., 2023b) clips with captions and 50 MUSDB18-HQ (Raffi et al., 2017) mixtures. For each clip we record one intermediate activation snapshot at the normalized rectified-flow time $t^* = 0.5$, halfway between data and noise, and keep the hidden states at the block indices $\ell \in \{6, 12, 18\}$. This yields $N \times T = 32,400$ token activations per studied layer.

Linear probing. We label clips against thirteen instrument and genre keyword lists provided in the supplementary material. For MUSICCAPS we match the captions, and for MUSDB18-HQ we derive instrument labels from the available stem metadata. We keep the eleven concepts $\mathcal{K}$ with at least five positive examples, excluding jazz and ambient. For each $k \in \mathcal{K}$ we time-pool the activations as

$\bar{h}_\ell(x) = \frac{1}{T} \sum_t H_\ell(x)[t, :]$ and fit a class-balanced logistic regression, reporting 5-fold stratified-CV AUC.

CAV steering. The fitted probe yields a coefficient vector $v_k \in \mathbb{R}^D$, which we ℓ_2 -normalize to $\tilde{v}_k$. To steer generation we hook DiT layer ℓ and additively shift its output,

$$h_\ell^{\text{steered}} = h_\ell + \alpha \cdot \tilde{v}_k, \quad (1)$$

broadcast across batch and time at every diffusion step. For each prompt–concept sweep we reuse the same random seed across all steering strengths, so changes are not caused by different initial noise draws.

Probe-score evaluation. We re-encode each generated audio $\tilde{x}$ through the same VAE and noising pipeline used to build the analysis corpus, read out the time-pooled hidden state $\bar{h}_\ell(\tilde{x})$ at the same $t^* = 0.5$ snapshot, and compute the probe score $s_k(\tilde{x}) = v_k^\top \bar{h}_\ell(\tilde{x}) + b_k$. The re-encoding uses a fixed neutral prompt with no steering applied so the score reflects audio content rather than the generation prompt. As a summary statistic we report the *slope per $\pm 30\alpha$* , the probe score at strength 30 minus the probe score at strength -30 . Since this readout shares its direction with the intervention, we complement it with the independent CLAP check below.

4. Results

Linear probing recovers concept directions. Multi-label logistic regressions on time-pooled activations cleanly separate musical concepts in Table 1, with instruments such as drums, bass, and electronic reaching AUC in the 0.85–0.92 range. Mean AUC is approximately constant across the three layers, indicating that concept information is broadly available throughout the depth of the DiT. We use the probe coefficient v_k as the concept direction in the steering experiments that follow.

CAV steering is monotonic on the probe metric. Treating v_k as a steering vector (Eq. 1) and varying $\alpha \in \{-30, -15, 0, +15, +30\}$ at layer 12, all five tested concepts produce monotonic probe-score responses (Figure 2). Sweeping α across ± 30 moves the probe score by several times the 5–10-unit clip-to-clip baseline variation, so the probe direction is not only predictive but causally effective in activation space. Whether this translates into perceptual change is taken up by the CLAP check below.

Steering is diagonal-dominant in concept space. Steering effects are diagonal-dominant rather than perfectly disentangled (Table 2). At a steering strength of 30 the mean on-diagonal change is 13.6 while the mean off-diagonal magnitude is 3.8, a ratio of 3.6, and every steering direction has its largest effect on its own concept. The nonzero

Table 1. Per-concept probe AUC across the three studied layers, with positive-clip count n_+ in the 150-clip analysis corpus. The best layer per concept is in bold. Instruments dominate, structural categories are noisier, and concept information is broadly available across depth.

Concept	n_+	L6	L12	L18
drums	52	0.92	0.86	0.87
bass	38	0.91	0.86	0.84
electronic	23	0.88	0.90	0.89
guitar	35	0.82	0.80	0.78
vocal_male	46	0.79	0.77	0.77
vocal_female	16	0.75	0.77	0.80
rock	16	0.77	0.78	0.67
classical	22	0.71	0.73	0.77
piano	27	0.60	0.67	0.62
strings	23	0.65	0.60	0.66
folk	22	0.57	0.53	0.50
mean		0.76	0.75	0.74

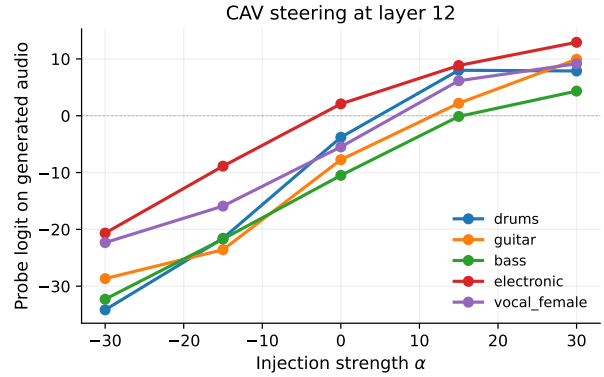


Figure 2. CAV steering curves at layer 12. Probe score on the generated audio responds monotonically to the injection strength α for all five concepts.

off-diagonal entries reflect expected musical entanglement among instruments and genres, but the diagonal dominance shows that the directions are far more specific than a single global “more music” axis.

Earlier layers are more controllable. Repeating the steering protocol at the early and late layers, layer 6 attains a mean slope of 24.0, more than thirty percent larger than the 18.2 at layer 12 and the 17.5 at layer 18. A probe-AUC sweep at every one of the 24 DiT layers, reported in the supplementary material, shows the same shape on the analysis side, with AUC peaking around layer 5 and plateauing thereafter. Early activations appear more linearly structured around the concept axes, and early interventions also propagate through more downstream blocks, so steering early is the more effective choice.

Table 2. Steering specificity at layer 12: change in probe score at steering strength 30 relative to baseline, for each steering direction (rows) measured on each concept probe (columns). On-diagonal entries (bold) dominate every row.

steer \ meas	drums	bass	elec	guit	voc
drums	+ 11.2	-1.5	+2.8	-0.8	-5.6
bass	+7.1	+ 14.2	-0.3	+5.0	+2.6
electronic	-0.7	-9.7	+ 10.5	-5.8	-4.6
guitar	+1.1	-1.3	-6.4	+ 17.3	+1.4
vocal_f	-5.1	+2.4	+6.7	-5.4	+ 14.5

CAVs transfer across prompts. Our main steering numbers use a single neutral prompt, so we test whether the directions transfer. We re-run the steering protocol with five musically distinct base prompts spanning jazz, rock, classical, ambient, and hip-hop, using layer-12 concept directions for drums, bass, and electronic, and report the full table in the supplementary material. All fifteen prompt-concept pairs yield positive Δ probe score, with per-concept means of 16.0 for bass, 12.1 for drums, and 16.0 for electronic. Where an effect is smaller, the base prompt already elicits the concept and leaves little headroom for the direction to add more. The directions therefore encode prompt-invariant concept axes rather than idiosyncratic features of any base distribution.

An independent acoustic check with CLAP. Because our probe-score readout shares its direction with the intervention, we add an external check that does not. We embed each steered generation with a frozen CLAP audio-language model (Wu et al., 2023) and measure cosine similarity between the audio and a short text prompt describing the target concept (full numbers in the supplementary material). The suppression direction is the most consistently corroborated: at a strength of -30 the concept-text similarity drops toward zero for all five concepts, far below the baseline range of 0.20 to 0.49, so the negative-strength effect we read from the probe is also detectable by an independent audio-language model rather than being an artefact of the shared readout. Positive steering is more sensitive to strength. In the cross-prompt setting, where the base prompt leaves room to add the concept, the CLAP change from baseline to a strength of 10 is positive for electronic (0.03, all five prompts) and bass (0.06), but when pushed to 30 the change turns negative for all three concepts even though the probe score keeps rising, suggesting that very strong injection moves activations off-distribution and begins to degrade the generation rather than adding clean concept content. CLAP is itself a learned proxy and we read these numbers as corroboration rather than ground truth; together with the probe-score response they suggest that probe-based steering acts as a usable suppression tool and additive knob at moderate strength, with the probe score overstating perceptual gains at off-distribution strengths.

5. Discussion

A matched sparse-autoencoder baseline (supplementary material) produced substantially weaker steering directions under the same activation budget, which is why the main text builds on probe-based directions. Applying the same pipeline to ACE-Step v1.5 turbo preserves probe decodability but loses the diagonal-dominant causal specificity reported above, suggesting that linear decodability alone is not sufficient for additive residual-stream control.

Limitations. Our 150-clip corpus is small, and a larger corpus might reveal finer or more numerous concept directions. Concept labels come from caption-keyword matching and are inherently noisy. We study Stable Audio Open 1.0; broader generalization to other music diffusion transformers such as DiffRhythm and newer Stable Audio variants remains open, as the ACE-Step check discussed above already suggests. Most importantly, the probe direction is used both as the intervention vector and as part of the readout, so probe-score changes confirm activation-space control but not perceptual change in the audio. We therefore complement it with an independent acoustic check using a frozen audio-language model and a small single-listener pilot reported in the supplementary material.

6. Conclusion

We presented a probing-based mechanistic interpretability and control study of a music diffusion transformer. Linear probes identify directions on Stable Audio Open’s intermediate activations that are both predictive and causally useful: injected additively into a single DiT layer, they shift probe-measured concept presence monotonically, act in a diagonal-dominant way across concepts, and survive changes in injection depth and base prompt. An independent audio-language model corroborates the suppression direction and suggests that moderate positive steering can add the target concept in some settings, while very strong steering leaves the data distribution and degrades the generation. The intervention is therefore best understood as a local steering knob within Stable Audio Open’s text-conditioned trajectory, not as a universal concept editor: a preliminary cross-model check on ACE-Step v1.5 turbo recovers probe decodability but loses causal specificity, suggesting that the property that makes probes useful here may reflect the specific way SAO routes concept information through the residual stream rather than a model-agnostic fact about music DiTs. Promising next steps include scaling the analysis corpus, a full multi-rater listening study, applying probe-based steering to music editing, and extending the cross-model panel.

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